



ST BENEDICT'S

SUBJECT	PHYSICAL SCIENCE PAPER 1
GRADE	11
EXAMINER	Mrs Govender
MODERATOR	Mr McCoy
DATE	18 July 2024
MARKS	165
DURATION	2,5 hours

%

LEARNER NAME: MEMO

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 24 pages and a separate data sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Additional space is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	9	10	Total
Mark											
Moderated Mark											
Question Total	14	20	14	20	18	11	15	14	24	12	165

Comment:

16 MARKS

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	C	D
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Here the option C has been marked as an example.

1.1	A ✓✓	B	C	D
1.2	A	B ✓✓	C	D
1.3	A	B ✓✓	C	D
1.4	A	B ✓✓	C	D
1.5	A	B ✓✓	C	D
1.6	A	B	C ✓✓	D
1.7	A	B ✓✓	C	D
1.8	A ✓✓	B	C	D

QUESTION 2 FMC**20 MARKS**

2.1 the rate of change of velocity. ✓ (1)

2.2 $v = u + at$ $m = \frac{20-13,5}{2-3,5}$

$13,5 ✓ = 20 + a (1,5) ✓$ OR $= - 4,33$

$a = - 4,33$ $= 4,33 \text{ m.s}^{-2} \text{ South}$

$a = 4,33 \text{ ms}^{-2} \text{ South} ✓$ (3)

2.3 $13,5 \times 3,6 ✓ = 48,6 \text{ km.h}^{-1} ✓$ OR $50 \div 3,6 = 13,88$

Yes he stops on time. $48,6$ is lower than $50 \text{ km.h}^{-1} ✓$

$13,5$ is lower than $13,88$

2.4 average speed = $\frac{\text{total distance}}{\text{total time}}$

$= \frac{65,125 ✓}{3,5 ✓}$

$= 18,61 \text{ m.s}^{-1} ✓$ (3)

2.5.1 $sc = sd ✓$

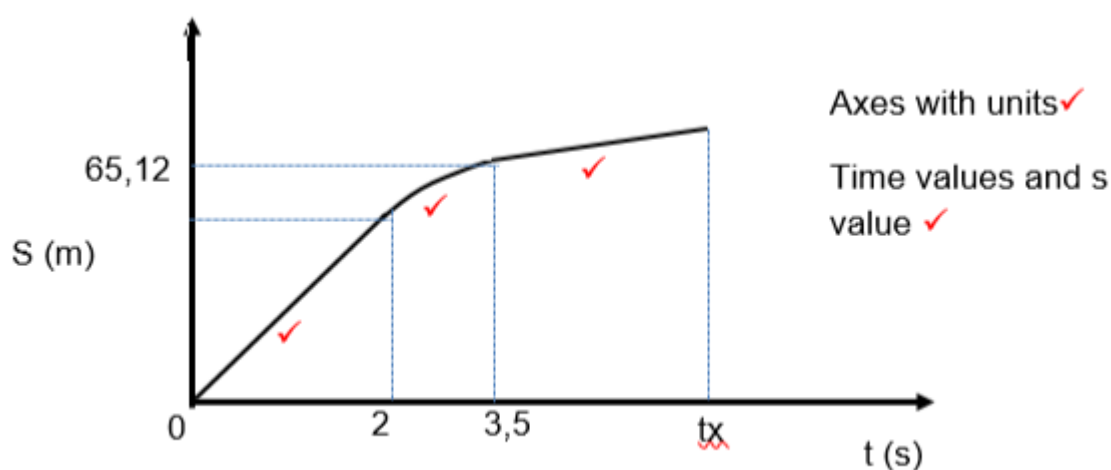
$\frac{1}{2} at^2 = v.t$

$0,5 \cdot 0,75t^2 ✓ = 13,5 t ✓$

$t = 36 \text{ s} ✓$

total time taken = $3,6 + 3,5 + 2,5 = 42 \text{ s} ✓$ (5)

2.5.2 (5)



QUESTION 3 KH**14 MARKS**

3.1.1 the gravitational force the Earth exerts on any object ✓ on or near its surface ✓ (2)

3.1.2 Action : Force of the Earth pulling down on the rocket. ✓

Reaction: Force of the rocket pulling up on the Earth. ✓

Action: Force of the gases pushing up on the rocket ✓

Reaction: Force of the rocket pushing down on the gases ✓ (2)

3.1.3 $F_{\text{net}} = ma$ ✓ (4)

Thrust – F_g – air resistance = ma

Thrust – $(0,45)(9,8) - 0,3$ ✓ = $(0,45)(7,5)$ ✓

Thrust = **8,09 N** ✓

3.1.4 $s = ut + \frac{1}{2}at^2$
 $= 0 + 0,5 \times 7,5 \times 3^2$ ✓
 $= \mathbf{33,75 \text{ m}}$ ✓ (2)

3.1.5 $v = u + at$
 $= 0 + (7,5)(3)$ ✓
 $= 22,5 \text{ m.s}^{-1} \text{ upwards}$ ✓

3.1.6 $F_{\text{net}} = F_g - F_f$
 $= 0,12(9,8) - 0,1$ ✓ **OR** $1,176 - 0,1$
 $= \mathbf{1,1 \text{ N downwards}}$ ✓ (2)

QUESTION 4 SG

19 MARKS

4.1

$$v = u + at \quad \checkmark$$

$$= 0 + (9,8) (0,64) \quad \checkmark$$

$$= 6,27 \text{ m.s}^{-1} \text{ down} \quad \checkmark$$

(3)

4.2

$$v = u + at$$

$$0 \quad \checkmark = u + (-9,8) \quad \checkmark \left(\frac{1,9 - 0,67}{2} \right) \quad \checkmark \text{ OR } (0,615)$$

$$= 6,03 \text{ m.s}^{-1} \quad \checkmark$$

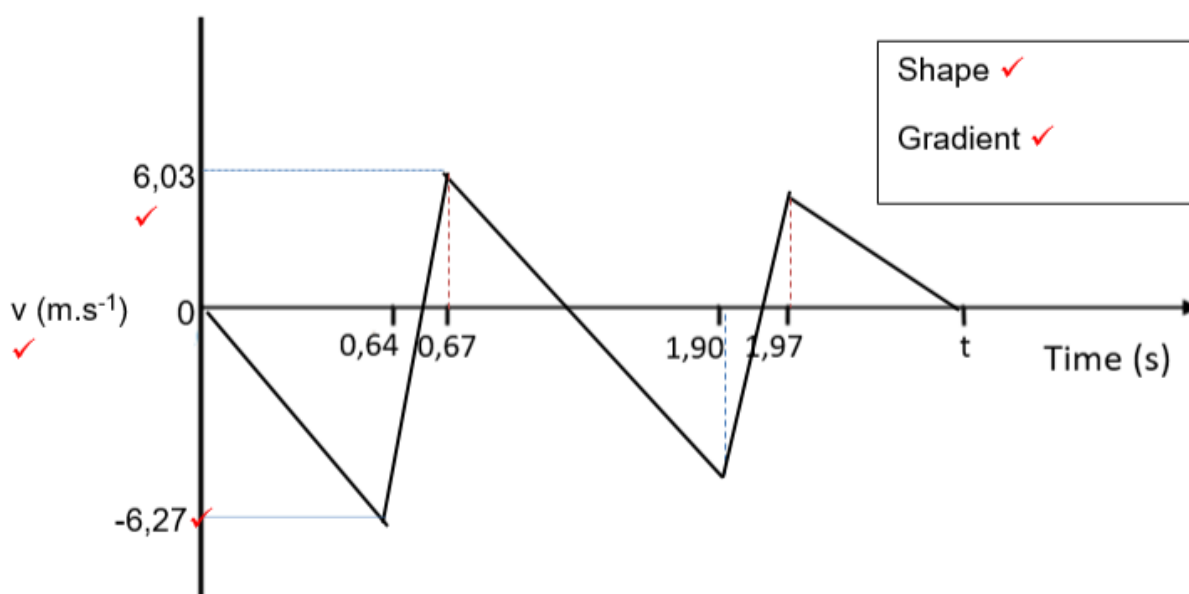
(4)

4.3

(5)

4.3

4.3



4.4

$$a = \frac{\Delta y}{\Delta x} \text{ (gradient)} \quad \checkmark$$

$$= \frac{6,03 - (-6,27)}{0,67 - 0,64} \quad \checkmark$$

6,03 coe from 4.2

$$= 410 \text{ m.s}^{-2} \quad \checkmark \text{ OR } 409,67 \text{ m.s}^{-2}$$

(3)

4.5

$$F_{\text{net}} = ma$$

$$F_{\text{court}} - F_g = ma$$

$$F_{\text{court}} - 0,04 (9,8) \quad \checkmark = 0,04 \quad \checkmark (410) \quad \checkmark \text{ c.o.e. from 4.4}$$

$$= 16,79 \text{ N upwards} \quad \checkmark$$

(4)

QUESTION 5 SG**20 MARKS**

5.1 1,37 ✓✓ (2)

5.2 when the only force acting on the object is the weight (gravitational force) ✓✓ (2)

5.3 Graph (6)

Scale on y axis , Scale on x axis	✓
First and last point plotted correctly	✓✓
Axes labelled with units Y – height (m) ,X – time ² (s ²)	✓✓
LOBF	✓

5.4 Gradient = $\frac{\Delta y}{\Delta x}$ (3)

= $\frac{0,2-1,2}{1,37-0,1}$ -1 if answer is positive

= -0,787 ✓ m.s⁻² ✓ shown on graph with dotted lines ✓

5.5 $s = ut + \frac{1}{2}gt^2$ gradient = $\frac{s}{t^2}$ ✓ (4)

$s = \frac{1}{2}gt^2$ ✓

$\frac{s}{t^2} = \frac{1}{2}g$ ✓

-0,787 = 0,5 g

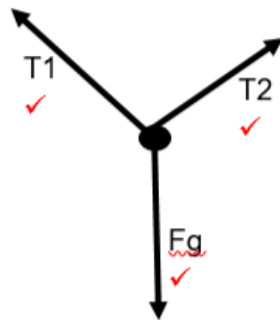
g = 1,57 m.s⁻² down ✓

5.6 1,25 – 1,33 m ✓✓ DO NOT ACCEPT 1,2 m

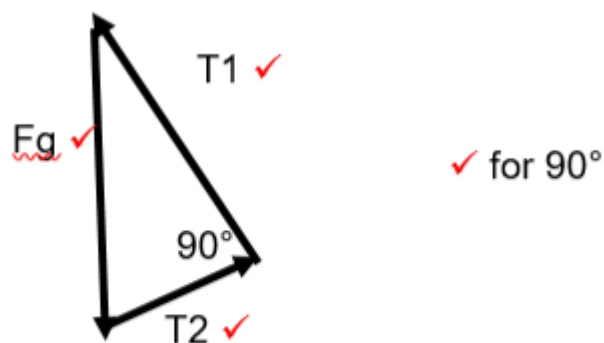
QUESTION 6 FMC**11 MARKS**

6.1 a physical quantity that has both magnitude and direction. ✓ (1)

6.2 (3)



6.3 (4)



6.4 (3)

$$\sin 45^\circ = \frac{4,9}{mg} \quad \checkmark$$

OR

$$F_g^2 = T_1^2 + T_2^2$$

$$mg = 6,93 \text{ N} \quad \checkmark$$

$$= 6,93 \quad \checkmark$$

$$m = 0,71 \text{ kg} \quad \checkmark$$

OR

$$F_2 = 0,5 \times 9,8$$

$$y_1 = F_2 \sin 45$$

$$= 4,9 \text{ N} \quad \checkmark$$

$$= 4,9 \sin 45 \quad \checkmark$$

$$= 3,46 \times 2$$

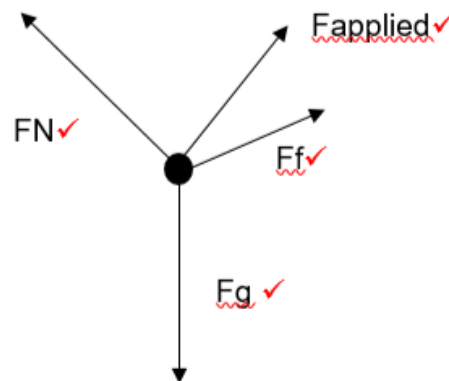
$$= 6,93 \div 9,8 = 0,7 \text{ kg} \quad \checkmark$$

QUESTION 7 KH

15 MARKS

7.1

(4)



7.2

$$\begin{aligned}
 F_{g\parallel} &= mg \sin \theta \checkmark \\
 &= 150 \times 9,8 \sin 35^\circ \checkmark \\
 &= \mathbf{843,16 \text{ N down the slope}} \checkmark
 \end{aligned}$$

(3)

7.3

(4)

$$\begin{aligned}
 F_f + F_x &= F_{g\parallel} \checkmark \\
 F_f + F \cos \theta &= \mathbf{843,16} \text{ coe from 7.2} \\
 F_f + 100 \cos 40^\circ &= 843,16 \checkmark \\
 F_f + 76.6 &= 843,16 \\
 \mathbf{F_f} &= \mathbf{766,56 \text{ N}} \checkmark
 \end{aligned}$$

$$7.4 \quad F_x + F_f - F_{g\parallel} = ma \checkmark$$

$$\begin{aligned}
 \mathbf{,766F} + \mathbf{842,911} - \mathbf{0,45 F} - \mathbf{843,16} &\text{coe from 7,2} = 0 \\
 0,316 F - 0,25 &= 0 \\
 \mathbf{F} &= \mathbf{0,79 \text{ N}} \checkmark
 \end{aligned}$$

$$\begin{aligned}
 F_x &= F \cos 40^\circ \\
 &= \mathbf{0,766 F} \checkmark
 \end{aligned}$$

(4)

$$F_{fs}^{max} = \mu_s F_N$$

$$\begin{aligned}
 &= 0,7 (FG - FV) \\
 &= 0,7 (mg \cos \theta - F \sin 40^\circ) \\
 &= 0,7 (150 \times 9,8 \cos 35^\circ - F \sin 40^\circ) \\
 &= 0,7 (1204,15 - 0,643F) \\
 &= \mathbf{842,91 - 0,45F} \checkmark
 \end{aligned}$$

QUESTION 8 ZN**14 MARKS**

- 8.1 When a net force, F_{net} , is applied to an object of mass, m , it accelerates in the direction of the net force. ✓ The acceleration, a , is directly proportional to the net force and inversely proportional to the mass. ✓ (2)

8.2 $F_{\text{net}} = 0$ ✓

$$T = \mu_s F_N$$

$$= 0,8 \times 9,8 \times 5,6 \text{ ✓}$$

$$= 43,9 \text{ N ✓}$$
 (3)

- 8.3 (5)

5,6 kg $F_{\text{net}} = ma$ $T - F_f = ma \text{ ✓}$ $T - 0,65 \times 5,6 \times 9,8 = 5,6 \times 0,25$ $T = 37,07 \text{ N ✓}$	M kg $F_{\text{net}} = ma$ $F_g - T = ma \text{ ✓}$ $M \times 9,8 - 0,25 M \text{ ✓} = 37,07$ $M = 3,88 \text{ kg ✓}$
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- 8.4 When object A exerts a force on object B, object B simultaneously ✓ exerts an oppositely directed force of equal magnitude on object A. ✓ (2)

- 8.5 Action: Normal force of table pushing the block up. ✓ (2)

Reaction: Force of the block pushing down on the table. ✓

OR

Action: Force of the Earth pulling down on the block ✓

Reaction: Force of the block pulling up on the Earth. ✓

QUESTION 9**24 MARKS**

9.1.1 resistance is a material's opposition to the flow of electric charge. ✓✓ (2)

9.1.2 (3)

$$P = \frac{V^2}{R} \quad \checkmark$$

$$R = \frac{12 \times 12}{24} \quad \checkmark$$

$$= 6 \, \Omega \quad \checkmark$$

9.2.1 40 J of energy is dissipated ✓ in 1 second. ✓ (2)

9.2.2 $R = \frac{V}{I} \quad \checkmark$ OR $P = I^2 R$

$$R = \frac{48}{0,8} \quad \checkmark$$

$$= 60 \, \Omega \quad \checkmark$$

$$R = \frac{P}{I^2} = \frac{40}{0,8^2}$$

$$= 62,5 \, \Omega$$

(3)

9.2.3 $V_3 = 54 \checkmark - 6 \checkmark = 48 \, V \quad \checkmark$ OR coe $V_3 = ,8 \times 62,5 = 50V$
(3)

$$V_3 = 0,8 \checkmark \times 60 \checkmark \text{ coe from 9.2.2} = 48 \, V \quad \checkmark$$

9.2.4 $R = \frac{V}{I}$

$$I = \frac{6}{1,5} \quad \checkmark$$

$$= 4 \, A \quad \checkmark$$

9.2.5 $IR = 4 - 0,8$ (3)

$$= 3,2 \, A \quad \checkmark$$

$$R = \frac{48}{3,2} \quad \checkmark \quad \text{coe from 9.2.3} = \frac{50}{3,2} = 20 \, V$$

$$= 15 \, \Omega \quad \checkmark$$

9.2.6 the rate of flow of charge ✓ (1)

9.2.7 DECREASES ✓✓ (2)

9.2.8

• The total resistance in the circuit increases. ✓ (3)

• $R = \frac{V}{I}$ V is constant ✓ and Current is inversely proportional to resistance. ✓

QUESTION 10**12 MARKS**

10.1.1 B ✓ Energy is inversely proportional to the wavelength. ✓ B has the smallest energy drop and the greatest wavelength. ✓ (3)

10.1.2 $\Delta E = 13,6 - 3,4$ ✓ (4)

$$= 10,2 \text{ eV}$$

$$E \text{ in J} = 10,2 \times 1,6 \times 10^{-19}$$

$$= 1,632 \times 10^{-18} \text{ J} \checkmark$$

$$E = hf \checkmark$$

$$1,632 \times 10^{-18} = 6,6 \times 10^{-34} \times f$$

$$f = 2,47 \times 10^{15} \text{ Hz} \checkmark$$

10.2.1 $c = f \times \lambda$ (3)

$$\lambda = \frac{3 \times 10^8 \checkmark}{7,81 \times 10^{19} \checkmark}$$

$$= 384,12 \text{ nm} \checkmark \text{ OR } 3,87 \times 10^{-7} \text{ m}$$

10.2.2 (2)

